

LAVJEET KUMAR RAI

☎ +91-8932018680 ✉ lavjeetkumarraai@gmail.com 💼 linkedin.com/in/lavjeet-rai 🐙 github.com/lavjeetrai

About

BTech student specializing in AI and Machine Learning focusing on predictive modeling and high-performance computing. Funded researcher working on neural network optimization for urban infrastructure. Experienced in developing real-time computer vision systems, fine-tuning LLMs, and building scalable full-stack pipelines to deploy AI solutions.

Education

SRM University, AP

Aug 2025 – June 2029

Bachelor of Technology – Computer Science (AI & ML Specialization), Amaravati, India

- Relevant Coursework: Data Structures, Algorithm Analysis, Database Management Systems, Computer Networking

Technical Skills

AI & Machine Learning: Supervised/Unsupervised Learning, LLM Fine-Tuning (PEFT, Ollama).

Languages: Python, C, C++, Java, JavaScript, Bash

Data & Tools: Pandas, NumPy, Matplotlib, Jupyter, Git/GitHub, Linux

Software Dev: Object-Oriented Programming, Data Structures & Algorithms, Next.js, Node.js, PostgreSQL, MongoDB

Research & AI Projects

Traffic Flow Prediction via RVFL Optimization

Funded University Research

- Secured a highly competitive INR 25,000 university seed grant to conduct independent AI research.
- Developing and optimizing Random Vector Functional Link (RVFL) neural networks to analyze and predict real-time urban traffic flow patterns specific to Amaravati's infrastructure.

Real-Time Indian Sign Language (ISL) Recognition

Computer Vision Project

- Engineered a real-time sign language translation system utilizing MediaPipe for precise hand-tracking.
- Implemented Long Short-Term Memory (LSTM) neural networks to accurately classify dynamic temporal gesture sequences.

LLM Fine-Tuning

Independent Research

- Learning Parameter-Efficient Fine-Tuning (PEFT) of Large Language Models using Ollama.

LearnX – AI Powered Smart Classroom

Live Project

- Developed Node.js backend services and designed scalable MongoDB schemas for real-time classroom data storage.
- Implemented role-based dashboards using React and Tailwind CSS, seamlessly integrating Google Gemini to power intelligent classroom features.

Accomplishments

- **Seed Grant Recipient:** Awarded INR 25,000 by SRM University-AP to fund applied machine learning research.
- **Hackathon Leader:** Led a team at the Google Cloud Gen AI Exchange Hackathon, driving the technical development of generative AI prototypes.
- **Lab Member:** Active researcher in the Singularity Student Research Lab, focusing on emerging tech.